SYMBOLIC RECURRENCE COEFFICIENTS FOR THE OSCILLATORY WEIGHT $x\,e^{im\pi x}$

KNOWLEDGE SENSE

ABSTRACT. Let m be a nonzero integer and $\eta = m\pi$. The monic polynomials orthogonal with respect to the bilinear form $\int_{-1}^{1} f(x)g(x)\,x\,e^{i\eta x}\,dx$ satisfy $xP_n = P_{n+1} + i\alpha_n P_n + \beta_n P_{n-1}$. Lyu and Zhou conjectured four closed formulas expressing $\alpha_{2k}, \alpha_{2k+1}$ for $k \geq 2$ and β_{2k}, β_{2k+1} for $k \geq 1$ through a single family of monic polynomials $G_j(\eta^2)$ of degree j. We prove this conjecture for every nonzero integer m. Regarded as polynomials in $z = \eta^2$, the denominator polynomials are normalizations of the moment Hankel determinants, and the numerator polynomials are normalizations of a combination of these determinants with their one-column bordered versions. Both determinants are n-fold integrals with polynomial amplitudes, and for such integrals the endpoint expansion terminates and is an exact finite identity. At integer frequencies, in addition, all vertex exponentials in this expansion take the same value. For the Hankel determinant this expansion gives the degree and the leading coefficient, in accordance with the large-frequency analysis of Celsus, Deaño, Huybrechs and Iserles. For the bordered determinant the dominant endpoint contribution vanishes, and we compute the first coefficient that survives. Together with the values of the normalized determinants at $z = 0$, this yields the numerator polynomials, and all four formulas then follow from one common family with rational coefficients that does not depend on m.

1. INTRODUCTION

Let m be a nonzero integer, put $\eta = m\pi$, and consider the complex weight

$$w(x) = x\,e^{i\eta x}, \qquad -1 \leq x \leq 1.$$

It defines the bilinear, non-Hermitian form $\langle f, g\rangle = \int_{-1}^{1} f(x)g(x)w(x)\,dx$ on polynomials. Since w is complex-valued, the existence theory for positive weights does not apply, and the existence of orthogonal polynomials is not automatic. Milovanović and Cvetković [4, Theorem 2.3] proved that for every nonzero integer m the monic orthogonal polynomials P_n, $\deg P_n = n$, exist for all $n \geq 0$. Because $\langle xf, g\rangle = \langle f, xg\rangle$, they satisfy a three-term

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recurrence, which we write as in [3, (1.5)] and [4, (3.1)]:

$$xP_n(x) = P_{n+1}(x) + i\alpha_n P_n(x) + \beta_n P_{n-1}(x), \qquad P_0 = 1, \quad P_{-1} = 0. \tag{1.1}$$

In [4] the first coefficients were computed symbolically and many more numerically, and a conjecture on their structure was formulated [4, Conjecture 3.1]. The closely related weight $e^{i\omega x}$ on $[-1, 1]$ was studied in connection with Gaussian quadrature for oscillatory integrals by Asheim, Deaño, Huybrechs and Wang [1], and the corresponding polynomials and Hankel determinants were analysed by Celsus, Deaño, Huybrechs and Iserles [2], whose large-frequency analysis of multiple oscillatory integrals plays a role below.

Recently Lyu and Zhou [3] derived, by ladder operators, coupled difference equations for the recurrence coefficients of the weights $x(1-x^2)^{\gamma-1/2}e^{i\eta x}$. For $\gamma = 1/2$ and $\eta = m\pi$ they obtained α_n and β_n for $n \le 9$ in closed form [3, Corollary 3.8, Table 4] and proposed the following conjecture.

Conjecture 1.1 ([3, Conjecture 3.9])**.** *There are monic polynomials G_j of degree j, with $G_0 = 1$ and $G_1(z) = z-2$, such that for every nonzero integer m and $\eta = m\pi$*

$$\alpha_{2k} = \frac{2(2k+1)}{\eta} + k(k+1)\,\eta\left(\frac{2G_{k^2-1}(\eta^2)}{G_{k^2}(\eta^2)} - \frac{G_{k(k+1)-1}(\eta^2)}{G_{k(k+1)}(\eta^2)}\right), \tag{1.2}$$

$$\alpha_{2k+1} = \frac{2(2k+2)}{\eta} + \eta\left(k(k+1)\frac{G_{k(k+1)-1}(\eta^2)}{G_{k(k+1)}(\eta^2)} - 2(k+1)(k+2)\frac{G_{k(k+2)}(\eta^2)}{G_{(k+1)^2}(\eta^2)}\right) \tag{1.3}$$

for $k \ge 2$, and

$$\beta_{2k} = -(2k)^2\,\frac{G_{k(k-1)}(\eta^2)\,G_{k(k+1)}(\eta^2)}{\eta^2\,G_{k^2}(\eta^2)^2}, \qquad \beta_{2k+1} = \frac{G_{k^2}(\eta^2)\,G_{(k+1)^2}(\eta^2)}{\eta^2\,G_{k(k+1)}(\eta^2)^2} \tag{1.4}$$

for $k \ge 1$.

The polynomials G_j for $j = 2, 3, 4, 5, 6, 8, 9, 11, 12, 15, 16$ are listed explicitly in [3, Appendix C(4)]. Our main result is the following.

Theorem 1.2. *There is a sequence $(G_j)_{j\ge0}$ of monic polynomials $G_j \in \mathbb{Q}[z]$ with $\deg G_j = j$, $G_0 = 1$ and $G_1(z) = z-2$, not depending on m, with the following property. For every nonzero integer m and $\eta = m\pi$ the numbers $G_{k^2}(\eta^2)$ and $G_{k(k+1)}(\eta^2)$, $k \ge 0$, are nonzero, the formulas* (1.2) *and* (1.3) *hold for all $k \ge 2$, and the formulas* (1.4) *hold for all $k \ge 1$. In particular, Conjecture 1.1 is true.*

The family is explicit. With $d_n = \lfloor n^2/4 \rfloor$, the polynomial G_{d_n} is a normalized Hankel determinant of the moments, and for $n \geq 4$ the polynomial G_{d_n-1} is a normalized combination of that determinant and of the corresponding one-column bordered determinant; see Section 6. The eleven polynomials listed in [3, Appendix C(4)] agree coefficient by coefficient with the members of this family (Section 7).

Determinants. Put $t = i\eta$ and $z = \eta^2$. After the removal of explicit powers of t, the Hankel determinant D_n of the moments and its one-column bordered version E_n (denoted Δ_n and Δ'_n in [4]; the prime does not denote a derivative) become polynomials $H_n(z)$ and $J_n(z)$ with integer coefficients that do not depend on m (Section 2). The recurrence coefficients are rational expressions in H_n and J_n, and Conjecture 1.1 becomes a statement about the degrees, the leading coefficients and the constant terms of these polynomials.

Endpoint expansion. Heine-type identities express D_n and E_n as n-fold integrals over $[-1,1]^n$ with the oscillator $e^{t(x_1+\cdots+x_n)}$. The expansion of such integrals in terms of the vertices of the cube is stated in [2, Proposition 3.2]; in one variable it is the steepest descent representation used in [4, Lemma 3.4] and [1, (5)]. For polynomial amplitudes this expansion terminates and is an exact identity. Moreover, at $t = im\pi$ every vertex exponential $e^{t(\varepsilon_1+\cdots+\varepsilon_n)}$, $\varepsilon \in \{\pm 1\}^n$, equals $(-1)^{mn}$. Grouping the vertices according to the number of coordinates equal to -1 gives an exact finite formula (Proposition 4.1). For H_n the dominant vertices yield the degree and the leading coefficient; this is the case $f(x) = \prod_a x_a$ of [2, Theorem 3.8]. For the bordered determinant the amplitude carries the additional factor $x_1 + \cdots + x_n$. For even n this factor vanishes at the dominant vertices, and for odd n the two dominant contributions cancel, so [2, Theorem 3.8] either does not apply or gives a vanishing leading term. The first coefficient that survives is determined in the following theorem.

Theorem 1.3. *For $n \geq 1$ put $d_n = \lfloor n^2/4 \rfloor$, $C_l = \prod_{j=0}^{l-1} (j!)^2$, and*

$$\lambda_{2k} = (-1)^k\, 4^{k^2} C_k^2, \qquad \lambda_{2k+1} = 2 \cdot 4^{k(k+1)} C_k C_{k+1},$$
$$\theta_{2k} = 2k^2, \qquad \theta_{2k+1} = (k+1)(3k+2).$$

Then H_n and J_n have degree d_n, the leading coefficient of H_n is λ_n, the leading coefficient of J_n is $\theta_n \lambda_n$, and $J_n(0) = n(n+1)H_n(0)$.

The assertion about H_n can also be read off from [2, Theorem 3.8] with $f(x) = \prod_a x_a$ (the factor $G(N+1)$ there is a Barnes function, and $G(l+1)^2 = C_l$), and the identity $J_n(0) = n(n+1)H_n(0)$ is [4, (2.8)]; we include short proofs of both. Thus the endpoint method, the constants C_l and the leading coefficients of H_n are known. What this paper adds is the leading coefficient $\theta_n \lambda_n$ of J_n, that is, the first surviving coefficient of the bordered determinant, and the simultaneous construction, described next, of a single family that gives all four formulas of Conjecture 1.1 in their full ranges.

Construction. For $n \geq 2$, Theorem 1.3 shows that the polynomial $n(n+1)H_n - J_n$ is divisible by z, has degree d_n, and has leading coefficient $(n(n+1) - \theta_n)\lambda_n \neq 0$. Normalizing H_n and $(n(n+1)H_n - J_n)/z$ to monic polynomials gives the denominators G_{d_n} and the numerators G_{d_n-1}. The two sets of indices do not overlap as soon as $n \geq 4$, which is exactly the range $k \geq 2$ of the formulas (1.2) and (1.3). The four formulas then reduce to identities between the normalizing constants.

Scope. To our knowledge, Conjecture 1.1 has not been proved before. We do not address the integrality assertions in [4, Conjecture 3.1], real frequencies that are not integer multiples of π, or the companion conjecture [3, Conjecture 3.7] for the weight $x(1-x^2)^{-1/2}e^{i\eta x}$.

Dependencies. The proof uses the existence theorem [4, Theorem 2.3], which guarantees that the recurrence coefficients are defined; see also Remark 6.2. It also uses the moment formula (2.1), which is [4, (2.5)], and the determinant criterion and formulas [4, Theorem 2.2, (3.2)] for orthogonal polynomials and their recurrence coefficients.

Organization. Section 2 expresses the recurrence coefficients through H_n and J_n. Section 3 collects the integral identities, and Section 4 proves the exact endpoint formula. Section 5 proves Theorem 1.3, and Section 6 constructs the family and proves Theorem 1.2. Section 7 reports the comparison with the polynomials of [3].

2. Moments and determinants

Throughout, m is a nonzero integer and

$$\eta = m\pi, \qquad t = i\eta, \qquad z = \eta^2 = -t^2, \qquad s = (-1)^m = e^t = e^{-t}.$$

The moments of w are $\mu_r = \int_{-1}^{1} x^r w(x)\,dx = \int_{-1}^{1} x^{r+1}e^{tx}\,dx$, $r \geq 0$. For $r \geq 0$ define

$$M_r(z) = (r+1)!\sum_{\ell=0}^{r} \frac{(1+(-1)^\ell)(-t)^\ell}{(\ell+1)!} = 2(r+1)!\sum_{j=0}^{\lfloor r/2\rfloor} \frac{(-z)^j}{(2j+1)!} \in \mathbb{Z}[z].$$

Thus $M_0 = 2$, $M_1 = 4$ and $M_2 = 12 - 2z$. In the present notation [4, (2.5)] reads

$$\mu_r = s\,(-1)^r\,t^{-r-1}M_r(z) \qquad (r \geq 0). \tag{2.1}$$

For $n \geq 1$ let

$$H_n(z) = \det\big[M_{a+b}(z)\big]_{0\leq a,b\leq n-1},$$

and let $J_n(z)$ be the determinant of the $n \times n$ matrix with entries $M_{a+b}(z)$, where $a = 0, \ldots, n-1$ and b runs through the column indices $0, 1, \ldots, n-2, n$. Thus $J_1 = M_1 = 4$ and $H_1 = M_0 = 2$. Put $H_0 = 1$ and $J_0 = 0$. Similarly, let D_n and E_n be the corresponding determinants built from the moments μ_{a+b}, with $D_0 = 1$ and $E_0 = 0$. These are the determinants Δ_n and Δ'_n of [4].

Lemma 2.1. *For $n \geq 0$,*

$$D_n = s^n\, t^{-n^2} H_n(z), \qquad E_n = -s^n\, t^{-n^2-1} J_n(z).$$

Proof. For $n = 0$ there is nothing to prove. By (2.1), the (a, b) entry is $s\,t^{-1}\big((-1)^a t^{-a}\big)\big((-1)^b t^{-b}\big)M_{a+b}$. Removing the factor $(-1)^a t^{-a}$ from row a, the factor $(-1)^b t^{-b}$ from column b, and $s\,t^{-1}$ from every row, we obtain $D_n = s^n t^{-n}(-1)^{2\sigma} t^{-2\sigma} H_n$ with $\sigma = n(n-1)/2$, which is the first identity. For E_n the sum of the column indices is $\sigma + 1$, so the same operations give the factor $s^n t^{-n}(-1)^{2\sigma+1} t^{-2\sigma-1} = -s^n t^{-n^2-1}$. □

Lemma 2.2. *For $n \geq 0$ the number $H_{n+1}(\eta^2)$ is nonzero, and*

$$\begin{aligned} \alpha_n &= \frac{1}{\eta}\left(\frac{J_{n+1}(\eta^2)}{H_{n+1}(\eta^2)} - \frac{J_n(\eta^2)}{H_n(\eta^2)}\right), \\ \beta_{n+1} &= -\frac{H_{n+2}(\eta^2)\, H_n(\eta^2)}{\eta^2\, H_{n+1}(\eta^2)^2}. \end{aligned} \tag{2.2}$$

Proof. By [4, Theorem 2.3] the orthogonal polynomials exist, which is equivalent to $D_n \neq 0$ for all $n \geq 1$ [4, Theorem 2.2]; by Lemma 2.1, $H_n(\eta^2) \neq 0$. The determinant formulas

$$i\alpha_n = \frac{E_{n+1}}{D_{n+1}} - \frac{E_n}{D_n}, \qquad \beta_n = \frac{D_{n+1} D_{n-1}}{D_n^2}$$

are [4, (3.2)]. For completeness: by Cramer's rule the coefficient of x^{n-1} in P_n is $-E_n/D_n$, and comparing the coefficients of x^n in (1.1) gives the first formula; the second follows from $\langle P_n, P_n\rangle = D_{n+1}/D_n$ and

$$\beta_n \langle P_{n-1}, P_{n-1}\rangle = \langle xP_n, P_{n-1}\rangle = \langle P_n, P_n\rangle.$$

Substituting Lemma 2.1, we get $E_n/D_n = -t^{-1} J_n/H_n$ and $D_{n+1}D_{n-1}/D_n^2 = t^{-2} H_{n+1}H_{n-1}/H_n^2$. Since $t = i\eta$ and $t^2 = -\eta^2$, this is (2.2). □

For example, $\alpha_0 = (J_1/H_1)/\eta = 2/\eta$ and $\beta_1 = -H_2 H_0/(\eta^2 H_1^2)$ with $H_2 = M_0 M_2 - M_1^2 = -4(z-2)$, so $\beta_1 = 1 - 2/\eta^2$, in accordance with [3, Table 4].

3. Integral identities

For $x = (x_1, \ldots, x_n)$ we write

$$V(x) = \prod_{1 \leq a < b \leq n} (x_b - x_a), \qquad p_1(x) = x_1 + \cdots + x_n,$$

with $V = 1$ if $n \leq 1$; thus p_1 is the first power sum.

Lemma 3.1 (Heine-type identities)**.** *Let ρ be a complex Borel measure on $\mathbb{R}$ with $\int |x|^k\, d|\rho|(x) < \infty$ for all $k \geq 0$, and let $\nu_k = \int x^k\, d\rho$. For $n \geq 1$,*

$$\det[\nu_{a+b}]_{0 \leq a,b \leq n-1} = \frac{1}{n!} \int_{\mathbb{R}^n} V(x)^2\, d\rho^{\otimes n}(x),$$

$$\det[\nu_{a+b}]_{0 \leq a \leq n-1,\ b \in \{0,\ldots,n-2,n\}} = \frac{1}{n!} \int_{\mathbb{R}^n} p_1(x)\, V(x)^2\, d\rho^{\otimes n}(x).$$

Proof. Let $0 \le \ell_1 < \cdots < \ell_n$ be integers. By multilinearity in the columns and Fubini's theorem,

$$\det\big[\nu_{a+\ell_c}\big]_{0\le a\le n-1,\,1\le c\le n} = \int \det\big[x_c^{a+\ell_c}\big]\, d\rho^{\otimes n}(x) = \int \prod_{c=1}^{n} x_c^{\ell_c}\, V(x)\, d\rho^{\otimes n}(x).$$

The measure $\rho^{\otimes n}$ is invariant under permutations of the coordinates and V is alternating, so averaging over all permutations replaces $\prod_c x_c^{\ell_c}$ by $(n!)^{-1}\det[x_c^{\ell_a}]_{a,c}$. For $(\ell_c) = (0,1,\dots,n-1)$ this alternant is $V(x)$. For $(\ell_c) = (0,1,\dots,n-2,n)$ it is an alternating polynomial, hence divisible by V, and the quotient is a symmetric polynomial that is homogeneous of degree one, hence equal to $\kappa\, p_1(x)$ for a constant κ. The monomial $x_n^n \prod_{j=1}^{n-1} x_j^{j-1}$ has coefficient 1 in the alternant; in $p_1(x)V(x)$ it arises only as x_n times the monomial $\prod_{j=1}^{n} x_j^{j-1}$ of V, which has coefficient 1, because for $j < n$ the exponent vector obtained by lowering the j-th exponent is not a permutation of $(0,\dots,n-1)$. Hence $\kappa = 1$, which also covers $n = 1$. □

Applied to $d\rho = x\, e^{tx}\, dx$ on $[-1,1]$, Lemma 3.1 gives, for $n \ge 1$,

$$\begin{aligned} D_n &= \frac{1}{n!}\int_{[-1,1]^n} F_0(x)\, e^{tp_1(x)}\, dx, \\ E_n &= \frac{1}{n!}\int_{[-1,1]^n} F_1(x)\, e^{tp_1(x)}\, dx, \end{aligned} \tag{3.1}$$

where $F_0(x) = V(x)^2 \prod_{a=1}^{n} x_a$ and $F_1(x) = p_1(x)F_0(x)$.

Lemma 3.2 (Exact endpoint representation)**.** *Let F be a polynomial in n variables with complex coefficients and let $t \in \mathbb{C}\setminus\{0\}$. Then*

$$\int_{[-1,1]^n} F(x)\, e^{tp_1(x)}\, dx = t^{-n} \sum_{\varepsilon\in\{\pm1\}^n} \Big(\prod_{a=1}^{n} \varepsilon_a\Big) e^{tp_1(\varepsilon)} \int_{(0,\infty)^n} F\big(\varepsilon - u/t\big)\, e^{-p_1(u)}\, du.$$

Proof. Both sides are linear in F, and for a monomial both sides factor into products of one-dimensional expressions. It therefore suffices to treat $n = 1$ and a polynomial f. Repeated integration by parts gives

$$\int_{-1}^{1} f(x) e^{tx}\, dx = \sum_{\varepsilon=\pm1} \varepsilon\, e^{\varepsilon t} \sum_{j\ge0} (-1)^j \frac{f^{(j)}(\varepsilon)}{t^{j+1}}.$$

On the other hand, Taylor's formula and $\int_0^\infty u^j e^{-u}\, du = j!$ give $\int_0^\infty f(\varepsilon - u/t)e^{-u}\, du = \sum_{j\ge0}(-1)^j f^{(j)}(\varepsilon)t^{-j}$. All sums are finite. □

For polynomial amplitudes Lemma 3.2 is thus an identity, not merely an asymptotic expansion. It is the terminating case of [2, Proposition 3.2], and its one-dimensional form is the representation behind [4, Lemma 3.4] and [1, (5)].

Lemma 3.3 (Laguerre blocks)**.** *For $l \geq 0$ and $\kappa \in \{0,1\}$ put $\Phi_{l,\kappa}(u) = \prod_{a=1}^{l} u_a^{\kappa}\, V(u)^2 e^{-p_1(u)}$ on $(0,\infty)^l$. Then*

(a) $\int_{(0,\infty)^l} \Phi_{l,0}(u)\,du = l!\,C_l$, *where $C_l = \prod_{j=0}^{l-1} (j!)^2$ and $C_0 = 1$;*
(b) $\int_{(0,\infty)^l} p_1(u)\,\Phi_{l,\kappa}(u)\,du = l(l+\kappa)\int_{(0,\infty)^l} \Phi_{l,\kappa}(u)\,du$.

Proof. Both statements are trivial for $l = 0$. (a) By Lemma 3.1 with $d\rho = e^{-u}du$ on $(0,\infty)$, the integral equals $l!\det[(a+b)!]_{0\le a,b\le l-1}$. Removing $a!$ from row a and $b!$ from column b leaves the matrix $\left[\binom{a+b}{a}\right]$, which equals $LL^{\mathsf T}$ with the unit lower triangular matrix $L = \left[\binom{a}{j}\right]_{a,j}$ by Vandermonde's identity $\binom{a+b}{a} = \sum_j \binom{a}{j}\binom{b}{j}$. Hence its determinant is 1.

(b) For $c > 0$ put

$$I(c) = \int_{(0,\infty)^l} \prod_{a} u_a^{\kappa}\, V(u)^2 e^{-c p_1(u)}\,du.$$

The substitution $u \mapsto u/c$ shows that $I(c) = c^{-l(l+\kappa)} I(1)$, since the integrand without the exponential is homogeneous of degree $l(l-1)+\kappa l$. For $c \in [1/2, 2]$ the derivative of the integrand in c is dominated by $p_1(u)\prod_a u_a^{\kappa} V(u)^2 e^{-p_1(u)/2}$, which is integrable. Differentiating at $c = 1$ gives $\int p_1(u)\,\Phi_{l,\kappa} = l(l+\kappa) I(1)$. □

4. The exact endpoint formula

For integers $r, q \geq 0$ let $\mathbb{E}_{r,q}$ denote expectation with respect to the probability measure on $(0,\infty)^r \times (0,\infty)^q$ with density

$$\frac{V(u)^2 V(v)^2 e^{-p_1(u)-p_1(v)}}{r!\,C_r\; q!\,C_q}, \qquad u \in (0,\infty)^r,\ v \in (0,\infty)^q,$$

which is a probability measure by Lemma 3.3(a). Define the polynomial

$$\mathcal{P}_{r,q}(h;u,v) = \prod_{a=1}^{r}(1+hu_a)\prod_{b=1}^{q}(1-hv_b)\prod_{a=1}^{r}\prod_{b=1}^{q}\Big(1+\frac{h(u_a-v_b)}{2}\Big)^2,$$

and the polynomials in h

$$W_{r,q}(h) = \mathbb{E}_{r,q}\big[\mathcal{P}_{r,q}(h;u,v)\big],$$
$$B_{r,q}(h) = \mathbb{E}_{r,q}\Big[\big(q-r-h(p_1(u)+p_1(v))\big)\mathcal{P}_{r,q}(h;u,v)\Big],$$

together with the constants

$$A_{r,q} = 4^{rq}\,C_r C_q, \qquad p_{r,q} = r^2 + q^2.$$

Proposition 4.1. *For $n \geq 1$ and every nonzero integer m,*

$$D_n = s^n \sum_{r=0}^{n} A_{r,n-r}\, t^{-p_{r,n-r}}\, W_{r,n-r}(1/t),$$

$$E_n = s^n \sum_{r=0}^{n} A_{r,n-r}\, t^{-p_{r,n-r}}\, B_{r,n-r}(1/t).$$

Proof. We apply Lemma 3.2 to (3.1). Let $\varepsilon \in \{\pm 1\}^n$ have r coordinates equal to -1 and put $q = n - r$. Then $e^{tp_1(\varepsilon)} = e^{t(q-r)} = s^{q-r} = s^n$, because $e^t = e^{-t} = s$ and $s^2 = 1$. Since F_0 and F_1 are symmetric, the integral attached to ε depends only on r, and each value of r occurs for $\binom{n}{r}$ vertices. We may therefore take $\varepsilon = (-1, \dots, -1, 1, \dots, 1)$ and write the integration variables as $(u, v) \in (0,\infty)^r \times (0,\infty)^q$, so that, with $h = 1/t$,

$$x_a = -1 - hu_a \quad (1 \leq a \leq r), \qquad x_{r+b} = 1 - hv_b \quad (1 \leq b \leq q).$$

Then $\prod_a x_a = (-1)^r \prod_a (1 + hu_a) \prod_b (1 - hv_b)$, and the sign $(-1)^r$ cancels the factor $\prod_a \varepsilon_a = (-1)^r$. For $a < a'$ and $b < b'$ the differences within the blocks are $x_{a'} - x_a = -h(u_{a'} - u_a)$ and $x_{r+b'} - x_{r+b} = -h(v_{b'} - v_b)$; the signs disappear after squaring, and these differences contribute $h^{r(r-1)+q(q-1)} V(u)^2 V(v)^2$ to $V(x)^2$. A difference between the blocks is $x_{r+b} - x_a = 2\big(1 + h(u_a - v_b)/2\big)$, and these contribute $4^{rq} \prod_{a,b} (1 + h(u_a - v_b)/2)^2$. Finally $p_1(x) = (q-r) - h(p_1(u) + p_1(v))$. Since $n + r(r-1) + q(q-1) = r^2 + q^2$, Lemma 3.2 gives

$$D_n = \frac{s^n}{n!} \sum_{r=0}^{n} \binom{n}{r} 4^{rq}\, t^{-(r^2+q^2)} \int V(u)^2 V(v)^2\, \mathcal{P}_{r,q}(h; u, v)\, e^{-p_1(u) - p_1(v)}\, du\, dv,$$

and the same expression with the extra factor $(q-r) - h(p_1(u) + p_1(v))$ for E_n. By the definition of $\mathbb{E}_{r,q}$ the integral equals $r!\, C_r\, q!\, C_q\, W_{r,q}(h)$, respectively $r!\, C_r\, q!\, C_q\, B_{r,q}(h)$, and $\binom{n}{r} r!\, q!/n! = 1$. □

In particular, at integer frequencies there is no relative oscillatory factor between the configurations with different r; this is the point where the hypothesis $\eta \in \pi\mathbb{Z}$ enters the endpoint analysis.

Lemma 4.2. *Let $r, q \geq 0$, $n = r + q$, and $b_{r,q} = (r - q)(rq + n)$. Then*

$$W_{r,q}(h) = 1 + b_{r,q}\, h + h^2\, \widetilde{W}_{r,q}(h),$$

$$B_{r,q}(h) = (q - r) + \big((q - r)\, b_{r,q} - (r^2 + q^2)\big)h + h^2\, \widetilde{B}_{r,q}(h),$$

with polynomials $\widetilde{W}_{r,q}, \widetilde{B}_{r,q}$.

Proof. The coefficient of h in $\mathcal{P}_{r,q}$ is $p_1(u) - p_1(v) + \sum_{a,b}(u_a - v_b) = (q + 1)p_1(u) - (r + 1)p_1(v)$. By Lemma 3.3(b) with $\kappa = 0$, $\mathbb{E}_{r,q}\, p_1(u) = r^2$ and $\mathbb{E}_{r,q}\, p_1(v) = q^2$. Hence the coefficient of h in $W_{r,q}$ is $(q + 1)r^2 - (r + 1)q^2 = (r - q)(rq + r + q) = b_{r,q}$. For $B_{r,q}$, multiply $\mathcal{P}_{r,q} = 1 + h\big((q + 1)p_1(u) - (r + 1)p_1(v)\big) + O(h^2)$ by $(q - r) - h(p_1(u) + p_1(v))$ and take expectations; the coefficient of h is $(q - r)b_{r,q} - (r^2 + q^2)$. □

5. Degrees and leading coefficients

In this section we prove Theorem 1.3. For $n \geq 1$ consider the Laurent polynomials in an indeterminate τ

$$\mathcal{L}_n(\tau) = \sum_{r=0}^{n} A_{r,n-r}\,\tau^{\,n^2-p_{r,n-r}}\,W_{r,n-r}(\tau^{-1}),$$
$$\mathcal{M}_n(\tau) = -\sum_{r=0}^{n} A_{r,n-r}\,\tau^{\,n^2+1-p_{r,n-r}}\,B_{r,n-r}(\tau^{-1}).$$

Lemma 5.1. *For $n \geq 1$, $H_n(-\tau^2) = \mathcal{L}_n(\tau)$ and $J_n(-\tau^2) = \mathcal{M}_n(\tau)$ in $\mathbb{C}[\tau,\tau^{-1}]$.*

Proof. By Lemma 2.1 and Proposition 4.1, $H_n(-t^2) = s^n t^{n^2} D_n = \mathcal{L}_n(t)$ and $J_n(-t^2) = -s^n t^{n^2+1} E_n = \mathcal{M}_n(t)$ for every $t \in i\pi\mathbb{Z}\backslash\{0\}$. After multiplication by a suitable power of τ, both sides of each identity become polynomials that agree at infinitely many points. □

Proof of Theorem 1.3. Write $q = n - r$. Since $p_{r,q} = \big(n^2 + (r-q)^2\big)/2$, the exponent $n^2 - p_{r,q} = \big(n^2 - (r-q)^2\big)/2$ is maximal exactly when $|r-q|$ is minimal.

Even $n = 2k$. The maximum $2k^2 = 2d_n$ is attained only for $r = q = k$, and every other r gives an exponent at most $2d_n - 2$. Since $W_{r,q}(0) = 1$, the coefficient of τ^{2d_n} in $\mathcal{L}_n$ is $A_{k,k} = 4^{k^2}C_k^2$, and no larger power occurs. In $\mathcal{M}_n$ the configuration $r = q = k$ contributes $-A_{k,k}\tau^{2d_n+1}B_{k,k}(\tau^{-1})$, and $B_{k,k}(h) = -2k^2h + O(h^2)$ by Lemma 4.2, since $q - r = 0$. Its top term is therefore $2k^2 A_{k,k}\tau^{2d_n}$, while all other configurations contribute powers at most τ^{2d_n-1}.

Odd $n = 2k+1$. The maximum $2k(k+1) = 2d_n$ is attained exactly for $(r,q) = (k,k+1)$ and $(k+1,k)$, both with $A_{r,q} = 4^{k(k+1)}C_kC_{k+1}$; every other r gives an exponent at most $2d_n - 4$. The coefficient of τ^{2d_n} in $\mathcal{L}_n$ is $2\cdot 4^{k(k+1)}C_kC_{k+1}$. In $\mathcal{M}_n$ the two dominant configurations contribute $\mp A_{r,q}\tau^{2d_n+1}$ from the constant terms $q - r = \pm 1$ of $B_{r,q}$, and these cancel. For the linear terms, $(q-r)b_{r,q} = -(q-r)^2(rq+n) = -(rq+n)$, so the coefficient of h in $B_{r,q}$ equals

$$-(rq+n)-(r^2+q^2) = -\big(k(k+1)+2k+1\big)-\big(2k^2+2k+1\big) = -(k+1)(3k+2)$$

for both configurations. Hence the top term of $\mathcal{M}_n$ is $2\theta_{2k+1}A_{k,k+1}\tau^{2d_n}$; the remaining configurations contribute powers at most τ^{2d_n-3}.

In both cases $\mathcal{L}_n(\tau) = (-1)^{d_n}\lambda_n\tau^{2d_n} + (\text{lower powers})$ and $\mathcal{M}_n(\tau) = (-1)^{d_n}\theta_n\lambda_n\tau^{2d_n} + (\text{lower powers})$; here we used $(-1)^{k^2} = (-1)^k$ and that $k(k+1)$ is even. If a polynomial $Q(z)$ has degree d and leading coefficient c, then $Q(-\tau^2)$ has top term $(-1)^d c\,\tau^{2d}$. By Lemma 5.1, H_n and J_n have degree d_n and leading coefficients λ_n and $\theta_n\lambda_n$.

It remains to compute the constant terms. Since $M_r(0) = 2(r+1)! = \int_0^\infty y^r \cdot 2y\, e^{-y}\, dy$, Lemma 3.1 with $d\rho = 2ye^{-y}dy$ on $(0,\infty)$ gives

$$H_n(0) = \frac{2^n}{n!}\int \Phi_{n,1}(y)\, dy, \qquad J_n(0) = \frac{2^n}{n!}\int p_1(y)\, \Phi_{n,1}(y)\, dy,$$

and Lemma 3.3(b) with $\kappa = 1$ yields $J_n(0) = n(n+1)H_n(0)$. □

Remark 5.2. The same argument shows $H_n(0) > 0$; its value $2^n \prod_{k=1}^{n-1} k!\,(k+1)!$ is computed in [4, (2.7)] through the Laguerre polynomials with parameter 1, and $J_n(0) = n(n+1)H_n(0)$ is [4, (2.8)].

6. Construction of the family and proof of Theorem 1.2

6.1. **Numerator polynomials.** For $n \geq 0$ define

$$K_n(z) = n(n+1)H_n(z) - J_n(z) \in \mathbb{Z}[z],$$

so that $K_0 = 0$ and $K_1 = 2 \cdot 2 - 4 = 0$. By Theorem 1.3, for $n \geq 1$ the polynomial K_n vanishes at $z = 0$, has degree at most d_n, and its coefficient of z^{d_n} is $e_n\lambda_n$, where

$$e_n = n(n+1) - \theta_n, \qquad e_{2k} = 2k(k+1), \qquad e_{2k+1} = k(k+1).$$

Indeed $2k(2k+1) - 2k^2 = 2k(k+1)$ and $(2k+1)(2k+2) - (k+1)(3k+2) = k(k+1)$. Thus $e_n \neq 0$ for $n \geq 2$, and

$$F_n(z) = \frac{K_n(z)}{e_n\lambda_n\, z} \qquad (n \geq 2)$$

is a monic polynomial in $\mathbb{Q}[z]$ of degree $d_n - 1$. For instance $F_2 = 1$ and $F_3 = z - 24$ (see Section 7).

6.2. **The family.** Define polynomials G_j, $j \geq 0$, by

$$G_{d_n} = \frac{H_n}{\lambda_n} \quad (n \geq 1), \qquad G_{d_n - 1} = F_n \quad (n \geq 4), \tag{6.1}$$

and $G_j = z^j$ for all other $j \geq 0$.

Lemma 6.1. *The definition* (6.1) *is consistent. Each G_j is a monic polynomial in $\mathbb{Q}[z]$ of degree j, independent of m, and $G_0 = 1$, $G_1 = z - 2$.*

Proof. We have $d_n - d_{n-1} = \lfloor n/2 \rfloor$ for $n \geq 1$, so $0 = d_1 < d_2 < d_3 < \cdots$, and $d_{n-1} < d_n - 1 < d_n$ for $n \geq 4$. Hence the indices d_n ($n \geq 1$) are distinct, the indices $d_n - 1$ ($n \geq 4$) are distinct, and no index of the second kind equals one of the first kind. Monicity and degrees follow from Theorem 1.3 and the definition of F_n, and $H_n, K_n \in \mathbb{Z}[z]$ do not involve m. Finally $G_0 = H_1/\lambda_1 = 2/2 = 1$, and $G_1 = H_2/\lambda_2 = -4(z-2)/(-4) = z - 2$. □

Note that the low indices $d_2 - 1 = 0$ and $d_3 - 1 = 1$ coincide with d_1 and d_2; this is why the numerators are taken only for $n \geq 4$. Since $F_3 = z - 24 \neq G_1$, the formulas (1.2) and (1.3) cannot be extended to $k = 1$ with the same family. In accordance with this, the factor $\eta^2 - 24$ appears in the entries for α_2 and α_3 in [3, Table 4].

6.3. **The recurrence coefficients.** For $n \geq 0$ let $R_n = K_n(\eta^2)/H_n(\eta^2)$, which is defined by Lemma 2.2. Since $J_n/H_n = n(n+1) - R_n$, Lemma 2.2 gives

$$\alpha_n = \frac{2(n+1) + R_n - R_{n+1}}{\eta} \qquad (n \geq 0), \tag{6.2}$$

and by (6.1), for $n \geq 4$,

$$R_n = e_n\,\eta^2\,\frac{G_{d_n-1}(\eta^2)}{G_{d_n}(\eta^2)}. \tag{6.3}$$

We also put $\lambda_0 = 1$, the value of the formula for λ_{2k} in Theorem 1.3 at $k = 0$; then $H_0 = \lambda_0 G_{d_0}$ with $d_0 = 0$. The normalizing constants satisfy, for $k \geq 1$ and $k \geq 0$ respectively,

$$\frac{\lambda_{2k+1}\lambda_{2k-1}}{\lambda_{2k}^2} = 4k^2, \qquad \frac{\lambda_{2k+2}\lambda_{2k}}{\lambda_{2k+1}^2} = -1. \tag{6.4}$$

Indeed, in the first ratio the signs cancel, the exponent of 4 is $k(k+1) + k(k-1) - 2k^2 = 0$, and the remaining factor is

$$\frac{4\,C_{k+1}C_{k-1}}{C_k^2} = \frac{4\,(k!)^2}{((k-1)!)^2} = 4k^2.$$

In the second ratio the sign is $(-1)^{k+1}(-1)^k = -1$, the exponent of 4 is $(k+1)^2 + k^2 - 2k(k+1) - 1 = 0$, and the factors C_k, C_{k+1} cancel.

Proof of Theorem 1.2. Let (G_j) be the family (6.1); by Lemma 6.1 it has the stated properties. Fix a nonzero integer m and put $\eta = m\pi$.

Denominators. For $k \geq 0$ we have $k^2 = d_{2k}$ and $k(k+1) = d_{2k+1}$. By Lemma 2.2, $G_{d_n}(\eta^2) = H_n(\eta^2)/\lambda_n \neq 0$ for $n \geq 1$, and $G_0 = 1$.

The formula for α_{2k}, $k \geq 2$. Take $n = 2k \geq 4$ in (6.2). By (6.3) with $d_{2k} = k^2$, $e_{2k} = 2k(k+1)$, $d_{2k+1} = k(k+1)$ and $e_{2k+1} = k(k+1)$,

$$\alpha_{2k} = \frac{2(2k+1)}{\eta} + 2k(k+1)\,\eta\,\frac{G_{k^2-1}(\eta^2)}{G_{k^2}(\eta^2)} - k(k+1)\,\eta\,\frac{G_{k(k+1)-1}(\eta^2)}{G_{k(k+1)}(\eta^2)},$$

which is (1.2).

The formula for α_{2k+1}, $k \geq 2$. Take $n = 2k+1 \geq 5$. Now $d_{2k+2} = (k+1)^2$, $d_{2k+2} - 1 = k(k+2)$ and $e_{2k+2} = 2(k+1)(k+2)$, so (6.2) and (6.3) give

$$\alpha_{2k+1} = \frac{2(2k+2)}{\eta} + k(k+1)\,\eta\,\frac{G_{k(k+1)-1}(\eta^2)}{G_{k(k+1)}(\eta^2)} - 2(k+1)(k+2)\,\eta\,\frac{G_{k(k+2)}(\eta^2)}{G_{(k+1)^2}(\eta^2)},$$

which is (1.3).

The formulas for β_n. By Lemma 2.2 and (6.1), for $n \geq 1$,

$$\beta_n = -\frac{\lambda_{n+1}\lambda_{n-1}}{\lambda_n^2} \cdot \frac{G_{d_{n+1}}(\eta^2)\,G_{d_{n-1}}(\eta^2)}{\eta^2\,G_{d_n}(\eta^2)^2},$$

where for $n = 1$ we use $H_0 = \lambda_0 G_{d_0}$. For $n = 2k$, $k \geq 1$, we have $(d_{2k-1}, d_{2k}, d_{2k+1}) = (k(k-1), k^2, k(k+1))$, and the first identity in (6.4) gives the first formula in (1.4). For $n = 2k+1$, $k \geq 1$, we have $(d_{2k}, d_{2k+1}, d_{2k+2}) = (k^2, k(k+1), (k+1)^2)$, and the second identity in (6.4) gives the second formula in (1.4). □

Remark 6.2. Lemma 2.2 used [4, Theorem 2.3] to ensure $H_n(\eta^2) \neq 0$. This can also be deduced from Theorem 1.3: H_n is a nonzero polynomial with rational coefficients, and $\eta^2 = m^2\pi^2$ is transcendental. This is essentially the argument of [4], where the nonvanishing of H_n is derived from $H_n(0) \neq 0$.

Remark 6.3. The argument treats positive and negative m simultaneously: (2.1), Lemma 2.1 and Proposition 4.1 hold for every nonzero integer m, and the family (6.1) depends only on $z = \eta^2$. This agrees with the reduction to $m > 0$ by complex conjugation in [4, Section 2]. For $n \leq 3$ the identities (6.2) and (2.2) reproduce the initial values in [3, Table 4].

Remark 6.4. The β-formulas alone determine the denominators. Indeed, (1.4) expresses β_n, $n \geq 2$, as a nonzero constant times $G_{d_{n+1}} G_{d_{n-1}}/(zG_{d_n}^2)$ evaluated at $z = \eta^2$. Suppose that another family of monic polynomials $\widetilde{G}_j$, not depending on m, with $\widetilde{G}_0 = 1$ and $\widetilde{G}_1 = z - 2$, satisfies (1.4), with nonzero denominators, for infinitely many m. Then $\widetilde{G}_{d_n} = G_{d_n}$ for all $n \geq 1$, by induction on n. Since $d_1 = 0$ and $d_2 = 1$, the cases $n = 1, 2$ are the initial values $\widetilde{G}_0 = G_0 = 1$ and $\widetilde{G}_1 = G_1 = z - 2$. Let $n \geq 2$ and assume $\widetilde{G}_{d_j} = G_{d_j}$ for $j \leq n$; the first step, $n = 2$, uses the formula for β_2 and yields $\widetilde{G}_{d_3} = G_{d_3}$, that is, $\widetilde{G}_2 = G_2$. Comparing the two expressions for β_n and cross-multiplying, we find that the polynomial $(\widetilde{G}_{d_{n+1}} - G_{d_{n+1}})\, G_{d_{n-1}}$ vanishes at $z = m^2\pi^2$ for infinitely many m, hence at infinitely many distinct points. It is therefore identically zero, and since $G_{d_{n-1}}$ is a nonzero polynomial, the polynomials $\widetilde{G}_{d_{n+1}}$ and $G_{d_{n+1}}$ coincide.

7. Exact computations

The proofs above do not depend on computer calculations. As a consistency check we computed H_n and J_n for $1 \leq n \leq 10$ in exact arithmetic. The entries M_r are integer polynomials, and both determinants were evaluated by fraction-free (Bareiss) elimination over $\mathbb{Z}[z]$, where every polynomial division was verified to be exact. From these determinants we formed $G_{d_n} = H_n/\lambda_n$ and, for $n \geq 4$, $G_{d_n - 1} = F_n$ with rational arithmetic. The computation confirms the degrees and leading coefficients of Theorem 1.3 and the identity $J_n(0) = n(n+1)H_n(0)$ for $n \leq 10$; see Table 1.

The resulting polynomials were compared with those listed in [3, Appendix C(4)]. For each of the eleven indices $j = 2, 3, 4, 5, 6, 8, 9, 11, 12, 15, 16$, the polynomial G_j of (6.1) coincides with the polynomial of [3]; altogether

TABLE 1. Degrees and leading coefficients for $1 \le n \le 10$.

n	d_n	λ_n	θ_n	e_n
1	0	2	2	0
2	1	−4	2	4
3	2	32	10	2
4	4	256	8	12
5	6	32768	24	6
6	9	−4194304	18	24
7	12	19327352832	44	12
8	16	89060441849856	32	40
9	20	26264993026824732672	70	20
10	25	−77458616235269355617648 64	50	60

102 coefficients agree. For example,

$$G_2 = z^2 - 13z + 6, \qquad G_3 = z^3 - 7z^2 + 12z - 360,$$
$$G_4 = z^4 - 14z^3 + 165z^2 - 1224z + 216, \qquad F_3 = z - 24,$$

and G_8 and G_{11} have non-integral coefficients, in agreement with the normalizations $2G_8$ and $2G_{11}$ used in [3, Appendix C(4)]. Here G_2, G_4 are denominators ($d_3 = 2$, $d_4 = 4$) and $G_3 = F_4$ is a numerator.

USE OF AI-ASSISTED TOOLS

AI-assisted tools were used in the research and preparation of this manuscript.

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Email address: `math@knowledgesense.jp`